



2487 - Detecting a Wide-Separation Gas Giant in AS 209

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
AS 209				
	1	AS 209 Roll 1	NIRCam Imaging	(1) EM-AS-209
	2	AS 209 Roll 2	NIRCam Imaging	(1) EM-AS-209

ABSTRACT

High angular resolution observations of planet forming disks are revealing a wide variety of disk substructures, predominantly azimuthally symmetric rings and gaps. Such morphologies can be readily explained as the secular outcome of planet-disk interactions, suggesting a large population of unseen planets. This scenario is also supported by kinematic studies of protoplanetary disks, which are revealing substantial deviations from Keplerian rotation characteristic of embedded giant planets. To test this scenario, a direct detection of an embedded planet is needed. We propose to use NIRCam to detect such a planet in the disk around AS 209. The circumstellar material exhibits a clear gap in the 12CO emission at a large orbital separation from the central star (210 au; 1.7"), indicating an extremely deep gap in the disk gas surface density. This is consistent with a

kinematic analysis showing a large pressure minimum at the center of the gap. The only mechanism known to produce such a feature in the gas surface density is a massive planet ($\sim 2 M_{\text{Jup}}$) opening a gap. The large gap depth and width and lower inclination of the disk are highly favorable conditions for detecting an embedded planet as extinction from disk material along the line of sight will be minimized. Using 3D numerical simulations of a bespoke AS 209 model as the basis for simulated observations, we find we will reach a 5 sigma sensitivity of $\sim 0.2 M_{\text{Jup}}$ at the location of the CO gap. As secondary goal, we will search for the planet generating the mm continuum rings at $0.8''$, where we obtain a 5 sigma sensitivity of $\sim 0.5\text{--}1 M_{\text{Jup}}$, far surpassing any limits achievable with ground-based facilities.

OBSERVING DESCRIPTION

We will use NIRCcam imaging to search for protoplanets that have been predicted to be present at a distance of 94 au ($0.78''$) and 210 au ($1.7''$) from the T Tauri star AS 209 (Zhang et al. 2018, Teague et al. 2018), with putative masses ranging from 0.2 to 2 Jupiter masses. We opt to use direct imaging rather than coronagraphic imaging because the inner planet is located near the inner working angle ($0.57''$) of the smallest mask available for the long wavelength channel. At $0.8''$, the fractional transmission of the coronagraphic mask is only 0.6, greatly reducing sensitivity to the potential inner protoplanet.

AS 209 will be observed at NIRCcam at two orientations separated by 10-14 degrees. The observations at the two orientations are linked in a non-interruptible sequence in order to minimize temporal PSF variations that would degrade the achievable contrasts. Roll subtraction is preferred over reference differential imaging with a separate PSF reference star because the circumstellar dust around AS 209 leads to an infrared excess that makes it difficult to find a star that would be suitable as a color match.

Each exposure will simultaneously observe with the F200W filter in the short wavelength channel and the F410M filter in the long wavelength channel. The F410M filter is selected for sensitivity to protoplanet emission, while the F200W filter will assist in rejecting background stars. We will use a subarray dither with 4 steps to help eliminate bad pixels. The fastest available readout pattern, RAPID, will be used to minimize saturation from the central star. Because of the cap of 200 integrations per exposure, we have subdivided the observations at each roll into separate exposures to achieve the required integration time. For faster readout times and lower data rates, we will also use the SUB160 subarray in Module B, which has a $5 \times 5''$ FOV in the short wavelength channel and $10 \times 10''$ FOV in the long wavelength channel. This is adequate to cover the entirety of the region where protoplanets are expected to be found around AS 209.

The total exposure time of 4000 s per roll was chosen to maximize the sensitivity at the predicted site of the inner protoplanet at $0.8''$.

Proposal 2487 - Targets - Detecting a Wide-Separation Gas Giant in AS 209

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	EM-AS-209	RA: 16 49 15.3036 (252.3137650d) Dec: -14 22 9.01 (-14.36917d) Equinox: J2000	Proper Motion RA: -7.417 mas/yr Proper Motion Dec: -23.733 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. The K-band magnitude is 6.961 +- 0.026. Category=Star Description=[Protoplanetary disks, T Tauri stars] Extended=YES				

Proposal 2487 - Observation 1 - Detecting a Wide-Separation Gas Giant in AS 209

Observation	Proposal 2487, Observation 1: AS 209 Roll 1										Thu Jul 21 22:00:30 GMT 2022
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	EM-AS-209	RA: 16 49 15.3036 (252.3137650d) Dec: -14 22 9.01 (-14.36917d) Equinox: J2000			Proper Motion RA: -7.417 mas/yr Proper Motion Dec: -23.733 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	The K-band magnitude is 6.961 +- 0.026.										
	Category=Star Description=[Protoplanetary disks, T Tauri stars] Extended=YES										
Template	Module					Subarray					
	B					SUB160P					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	SUBARRAY_DITHER		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F410M	RAPID	10	160	640	4	1964.902		
	2	F200W	F410M	RAPID	10	160	640	4	1964.902		
Special Requirements	Group Observations 1, 2, Non-interruptible Aperture PA Offset 1 from 2 by 10 to 14 Degrees (Same offsets in V3)										

Proposal 2487 - Observation 2 - Detecting a Wide-Separation Gas Giant in AS 209

Observation	Proposal 2487, Observation 2: AS 209 Roll 2										Thu Jul 21 22:00:30 GMT 2022
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	EM-AS-209	RA: 16 49 15.3036 (252.3137650d) Dec: -14 22 9.01 (-14.36917d) Equinox: J2000			Proper Motion RA: -7.417 mas/yr Proper Motion Dec: -23.733 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	The K-band magnitude is 6.961 +- 0.026.										
	Category=Star Description=[Protoplanetary disks, T Tauri stars] Extended=YES										
Template	Module					Subarray					
	B					SUB160P					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	SUBARRAY_DITHER		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F200W	F410M	RAPID	10	160	640	4	1964.902		
	2	F200W	F410M	RAPID	10	160	640	4	1964.902		
Special Requirements	Group Observations 1, 2, Non-interruptible Aperture PA Offset 1 from 2 by 10 to 14 Degrees (Same offsets in V3)										